

EXPERIMENT 9

Shravanya Andhale
D20A-02

AIM: Create an ERC 20 Token in Remix IDE

THEORY:

1.1 What is a Token?

A **token** is a digital asset created and managed using a blockchain's smart-contract functionality. Unlike the native cryptocurrency of a blockchain, a token is generally created by a smart contract according to a predefined token standard.

On Ethereum, tokens can follow standards such as:

- **ERC-20** – fungible tokens
- **ERC-721** – non-fungible tokens (NFTs)
- **ERC-1155** – supports both fungible and non-fungible assets

An ERC-20 token represents a **fungible asset**, meaning every unit of the token is interchangeable with another unit of the same token.

For example:

1 ABC token = 1 ABC token

Each token has the same value and properties as another token of the same type.

1.2 What is ERC-20?

ERC-20 stands for **Ethereum Request for Comment 20**. It defines a standard interface for fungible tokens on Ethereum-compatible blockchains.

The standard specifies functions and events that allow wallets, exchanges and decentralized applications to interact with tokens in a common way.

An ERC-20 token maintains information such as:

- Token name
- Token symbol
- Total supply
- Balances of individual addresses
- Allowances given to other addresses

1.3 Remix IDE

Remix IDE is a browser-based development environment used for writing, compiling, deploying and testing Ethereum smart contracts written in Solidity.

The major components used in this experiment are:

Solidity → Remix Compiler → Remix Deploy & Run → MetaMask → Sepolia → Ethereum Blockchain

Remix allows us to:

1. Write Solidity code.
2. Compile the smart contract.
3. Select an environment for deployment.
4. Connect to MetaMask.
5. Deploy the contract.
6. Execute contract functions.
7. View transaction results.

1.4 MetaMask

MetaMask is a cryptocurrency wallet that manages Ethereum accounts and allows users to interact with decentralized applications and smart contracts.

In this experiment, MetaMask is used to:

- Create/manage the Ethereum account.
- Connect Remix with the wallet.
- Select the Sepolia test network.
- Hold test ETH.
- Approve deployment transactions.
- Pay gas fees.
- Interact with the deployed ERC-20 contract.

1.5 Sepolia Testnet

Sepolia is an Ethereum test network used for development and testing.

Instead of real ETH, the network uses **test ETH**, which has no real monetary value.

Test ETH is required because deploying and interacting with a smart contract consumes **gas**.

The basic flow is:

Sepolia Faucet → Test ETH → MetaMask → Remix → Smart Contract Deployment

2. ERC-20 Token Functions

The important ERC-20 functions are:

Function	Purpose
name()	Returns the name of the token
symbol()	Returns the token symbol
decimals()	Returns the number of decimal places
totalSupply()	Returns total token supply
balanceOf(address)	Returns the token balance of an address
transfer(address, amount)	Transfers tokens to another address
approve(address, amount)	Gives another address permission to spend tokens
allowance(owner, spender)	Returns the amount a spender is allowed to spend
transferFrom(owner, recipient, amount)	Transfers tokens using an approved allowance

Example

Suppose:

Token = ABC

Total Supply = 1,000,000 ABC

If Alice approves Bob to spend 100 ABC:

Alice → approve(Bob, 100)

Bob can then use:

transferFrom(Alice, anotherAddress, 100)

subject to the available allowance.

3. ERC-20 Events

Events are emitted by smart contracts to record important activities that can be observed by blockchain applications.

Transfer Event

event Transfer(address indexed from, address indexed to, uint256 value);

It is emitted whenever tokens are transferred.

Example:

Alice → Bob : 100 ABC

generates a Transfer event.

Approval Event

```
event Approval(  
    address indexed owner,  
    address indexed spender,  
    uint256 value  
);
```

It is emitted when a token owner approves another address to spend tokens.

4. ERC-20 Attributes

Important ERC-20 token attributes include:

Name

Human-readable name of the token.

Example:

College Token

Symbol

Short representation of the token.

Example:

CLG

Decimals

Specifies how many decimal places the token supports.

The commonly used value is:

18

Total Supply

The total number of tokens currently existing.

Balance

The number of tokens held by a particular Ethereum address.

5. ERC-20 vs ERC-721

Feature	ERC-20	ERC-721
Type	Fungible	Non-fungible
Interchangeability	Tokens are interchangeable	Each token is unique
Example	USDT, DAI	NFTs
Token identity	Units are generally identical	Each token has a unique ID
Divisibility	Can represent fractional units	Individual NFT is generally treated as a unique asset
Main use	Currency, utility/governance tokens	Digital art, collectibles, ownership
Balance	Address can hold a quantity	Ownership is associated with specific token IDs

Mintable

Minting means creating new tokens.

Existing Supply = 1,000

Mint 500

New Supply = 1,500

It increases the total token supply.

Burnable

Burning means permanently destroying tokens.

Existing Supply = 1,500

Burn 200

New Supply = 1,300

It decreases the total token supply.

Pausable

The **Pausable** feature allows authorized accounts to temporarily stop token transfers or other selected token operations.

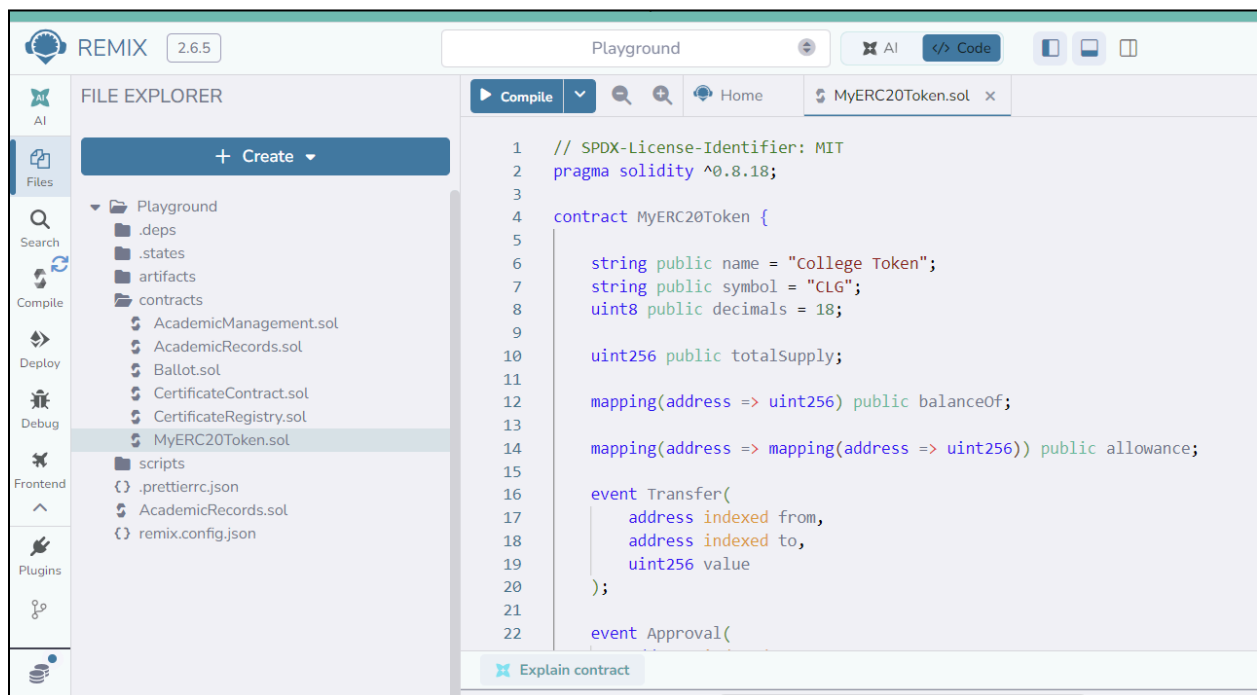
It can be useful in emergency situations such as:

- Security vulnerability
- Suspicious activity
- Smart-contract issue

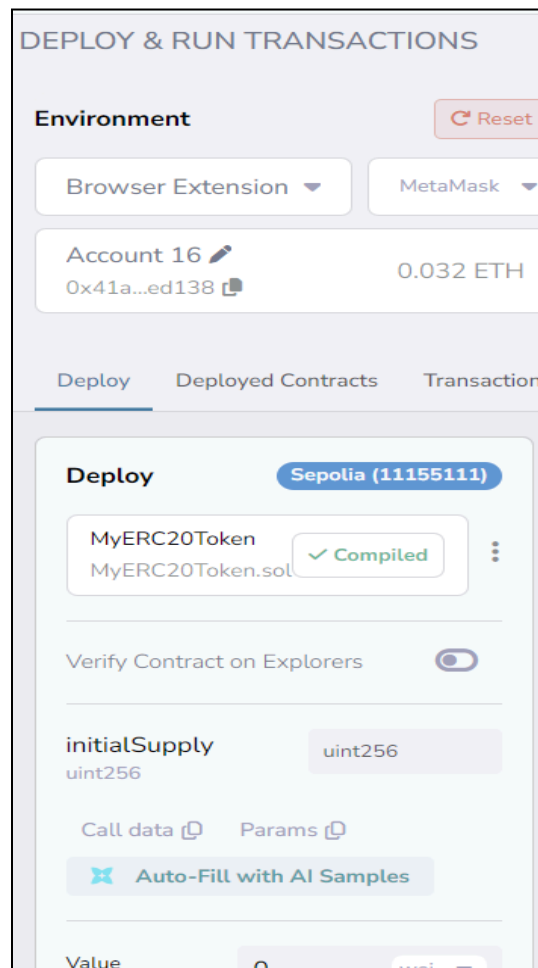
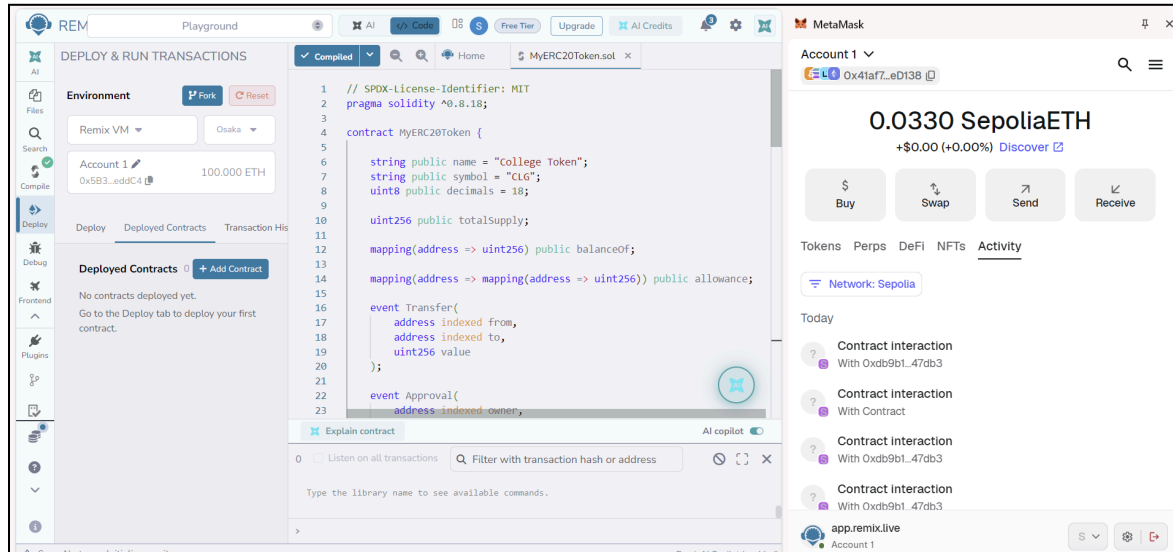
Once the contract is unpaused, normal operations can continue.

CODE AND OUTPUT:

Creating contract, compiling:



Connecting Metamask:



Deploying contract, making calls:

DEPLOY & RUN TRANSACTIONS

Environment

Reset

Browser Extension

MetaMask

Account 16

0x41a...ed138

0.032 ETH

Deploy

Deployed Contracts

Transaction His

Verify Contract on Explorers

initialSupply

uint256

1000000

Call data

Params

Auto-Fill with AI Samples

Value

0

wei

Gas limit

auto

0

Deploy

1 // SPDX-License-Identifier: MIT

2 pragma solidity ^0.8.18;

3

4 contract MyERC20Token {

5

6 string public name = "College Token";

7 string public symbol = "CLG";

8 uint8 public decimals = 18;

9

10 uint256 public totalSupply;

11

12 mapping(address => uint256) public balanceOf;

13

14 mapping(address => mapping(address => uint256)) public allowance;

15

16 event Transfer(

17 address indexed from,

18 address indexed to,

19 uint256 value

20);

21

22 event Approval(

23 address indexed owner,

24

Explain contract

AI copilot

0

Listen on all transactions

Filter with transaction hash or address

0

×

error: enclosing arguments: type(uint) address argument: string; empty string (arguments) value;

Account 1

Wallet 1

Deploy a contract

This site wants you to deploy a contract

Estimated changes

Unavailable

Network

s Sepolia

Request from

app.remix.live

Network fee

0.0092 SepoliaETH

Speed

Market ~12 sec

Cancel

Confirm

Transaction Recorder 3

Save

Contract Call

Transaction Lis

Newest

MyERC20Token

1 transaction

0xA60...d20bD

Deployed 2min ago

✓ constructor

tx: 0xf19...694fa

deploy

2min ago

MyERC20Token

1 transaction

0x101...2bBa7

Deployed 2min ago

Deployed Contracts 1

+ Add Contract

Interact with a deployed contract

MyERC20Token

Sepolia

0xA60...d20bD

2s ago

Functions

Search functions...

allowan... (address, addres...

Call

param0 (address)

param1 (address)

name

Call

0:string: College Token

symbol

Call

totalSupply

Call

transfer (address, uint256)

to (address)

Cannot read properties of undefined (reading 'result')

If the transaction failed for not having enough gas, try increasing the gas limit gently.

CA

LL

[call] from: 0x41af7ccae56dbff8ff51ff4192441966267ed138 to: MyERC20Token.name() data: 0x06f...dde03

Debug

symbol

Call

0:string: CLG

totalSupply

Call

transfer (address, uint256)

to (address)

If the transaction failed for not having enough gas, try increasing the gas limit gently.

CA

LL

[call] from: 0x41af7ccae56dbff8ff51ff4192441966267ed138 to: MyERC20Token.name() data: 0x06f...dde03

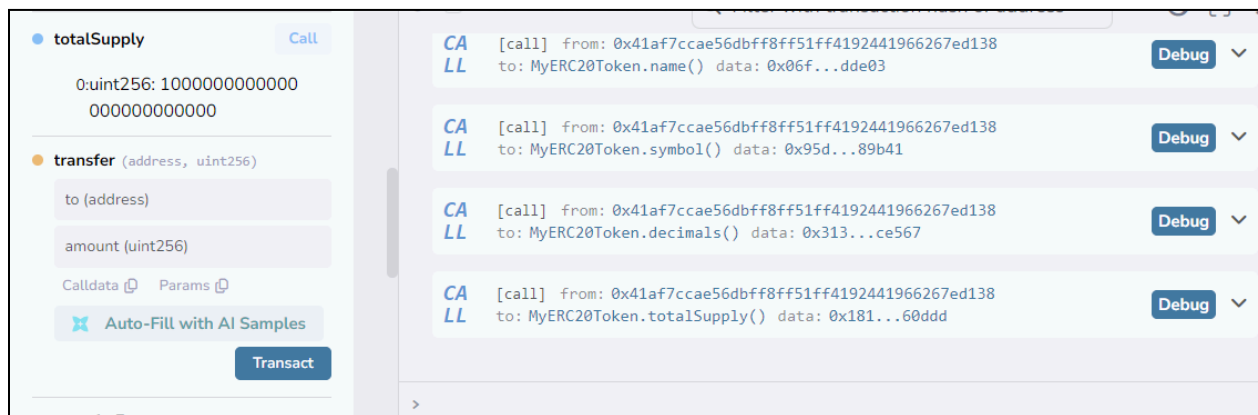
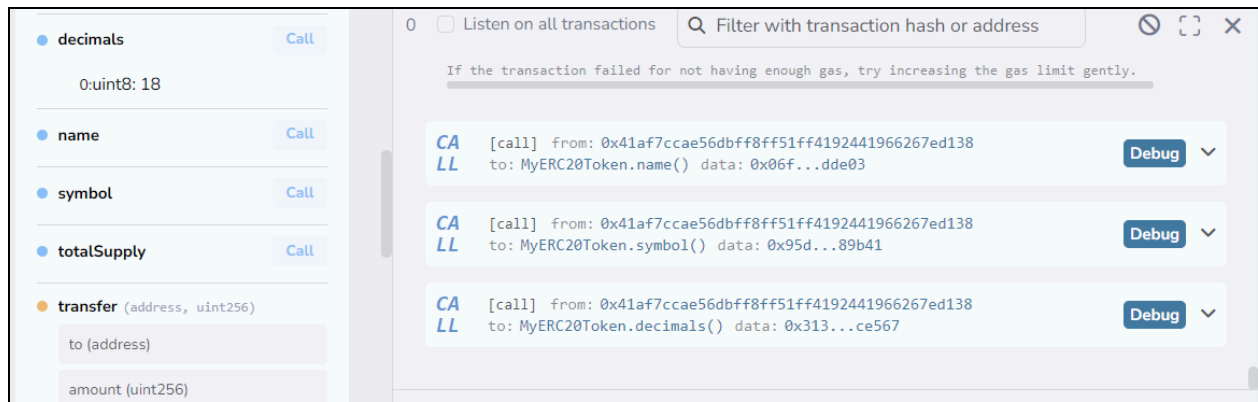
Debug

CA

LL

[call] from: 0x41af7ccae56dbff8ff51ff4192441966267ed138 to: MyERC20Token.symbol() data: 0x95d...89b41

Debug



Transferring to another Account:

